Algebra 1: Polynomials
Foundation Level

Name: ____________________
Date: ____________________

Question 1: Simplify the expression $x^2 + 3x + 2$ by factoring.

A) $x(x + 1)$
B) $x(x + 2)$
C) $(x + 1)(x + 2)$
D) $(x - 1)(x - 2)$
E) $(x + 1)(x - 2)$

Question 2: Find the value of x for which $2x^2 - 5x - 3 = 0$.

A) $x = -$
rac12orx = 3

A) $x = -$
rac32orx = 1

A) $x = -1$ or $x =$
rac32

A) $x =$
rac12orx = -3

A) $x = 1$ or $x = -1$

Question 3: What is the degree of the polynomial $3x^4 - 2x^2 + 1$?

A) 2
B) 3
C) 4
D) 5
E) 6

Question 4: Simplify $(x^2 + 2x + 1) + (x^2 - 2x - 1)$.

A) $2x^2$
B) $2x^2 + 2$
C) $2x^2 - 2$
D) $2x^2 + 4x$
E) $2x^2 - 4x$

Question 5: Factor out the greatest common factor from $12x^3 + 6x^2$.

A) $6x^2(2x+1)$
B) $6x(2x^2+1)$
C) $6(2x^3+x^2)$
D) $6x^2(2x-1)$
E) $3x(4x^2+2x)$

Question 6: Find the product of $(x+2)$ and $(x-2)$.

A) x^2+4
B) x^2-4
C) x^2-2x-4
D) x^2+2x-4
E) x^2-2x+4

Question 7: What is the value of the polynomial x^2+2x+1 when $x=-1$?

A) 1
B) 2
C) 3
D) 0
E) -1

Question 8: Simplify $(2x^2+3x-1)-(x^2-2x-1)$.

A) x^2+5x
B) x^2+x
C) x^2-x
D) x^2+5x-2
E) x^2+5x+2

Question 9: Factor the polynomial $x^3+2x^2-7x-12$. Show all work and explain your steps.

Question 10: Solve the equation $x^2+4x+4=0$. Show all steps and explain the method used.

Chef's Tips (Hints):

- When factoring or simplifying expressions, always look for common factors first.
- To solve quadratic equations, try factoring or using the quadratic formula $x =$ rac-b pm sqrtb$^2 - 4ac2a$.
- When simplifying or factoring, it's helpful to check your work by plugging in values for x to ensure correctness.